# Copulas: An Introduction

2026-04-17

## Introduction

A copula is a multivariate distribution with uniform marginals. It captures the dependence structure of a joint distribution independently of the marginals. Sklar's theorem says every joint distribution decomposes into its marginals plus a unique copula; this decomposition is the cleanest way to model dependence when the marginals are non-normal.

## Prerequisites

Joint and marginal distributions; cumulative distribution function; basic multivariate probability.

## Theory

**Sklar's theorem** (1959). For any joint CDF $F_{XY}$ with marginals $F_X, F_Y$, there exists a copula $C : [0,1]^2 \to [0,1]$ such that

$$F_{XY}(x, y) = C(F_X(x), F_Y(y)).$$

If $F_X, F_Y$ are continuous, $C$ is unique. In the other direction, for any copula $C$ and marginals $F_X, F_Y$, the composition defines a valid joint CDF.

**Common bivariate copulas**:

- **Gaussian copula**: induced by a bivariate normal with correlation $\rho$. Symmetric, no tail dependence.
- **t-copula**: from a bivariate t. Same symmetric structure as Gaussian but with heavier tail dependence.
- **Clayton copula**: asymmetric lower-tail dependence.
- **Gumbel copula**: asymmetric upper-tail dependence.
- **Frank copula**: symmetric, no tail dependence, but wider range of dependence than Gaussian.

**Tail dependence**: the lower and upper tail-dependence coefficients $\lambda_L = \lim_{u \to 0^+} C(u, u)/u$ and $\lambda_U = \lim_{u \to 1^-} (1 - 2u + C(u, u))/(1 - u)$ quantify how strongly extreme values of one variable co-occur with extremes of the other.

The **probability integral transform** turns any continuous marginal into a uniform: $U = F_X(X) \sim \text{Uniform}(0, 1)$. Copulas operate on the uniform scale and transform back to the original scale via $F_X^{-1}$.

## Assumptions

Continuous marginals (for uniqueness); otherwise the copula is unique only up to the joint probabilities of atoms.

## R Implementation

```r
library(copula)

# Fit a Gaussian copula to bivariate data
set.seed(2026)
```

```r
n <- 1000

# Simulate data with non-normal marginals but Gaussian copula
cop <- normalCopula(param = 0.7, dim = 2)
mvd <- mvdc(copula = cop,
            margins = c("gamma", "exp"),
            paramMargins = list(list(shape = 3, rate = 0.8),
                                list(rate = 0.3)))
X <- rMvdc(n, mvd)

# Visualise the dependence through the copula (uniform-scaled)
U <- pobs(X)   # empirical pseudo-observations on [0, 1]^2
plot(U, pch = 16, cex = 0.3, col = "#2A9D8F",
     main = "Empirical copula (pseudo-observations)",
     xlab = "U1", ylab = "U2")

# Fit a Gaussian copula
gc_fit <- fitCopula(normalCopula(dim = 2), data = U, method = "ml")
coef(gc_fit)

# Compare Clayton (lower tail dep) vs Gumbel (upper tail dep)
cc_fit <- fitCopula(claytonCopula(dim = 2), data = U, method = "ml")
gu_fit <- fitCopula(gumbelCopula(dim = 2), data = U, method = "ml")
c(gaussian = AIC(gc_fit), clayton = AIC(cc_fit), gumbel = AIC(gu_fit))
```

## Output & Results

```
[1] 0.7036       # MLE of copula correlation recovers the true 0.7

gaussian  clayton  gumbel
  -822.1   -551.0  -683.8
```

Gaussian copula has the best AIC, matching the data-generating model. The Clayton and Gumbel copulas mis-specify the tail structure and fit worse.

## Interpretation

Copula models are standard in finance (risk aggregation) and increasingly common in epidemiology (joint survival) and genetics (multivariate phenotypes). They let analysts specify marginals appropriate to each variable (gamma for concentrations, Beta for proportions) without forcing joint normality.

## Practical Tips

- Do not fit copulas without first checking that the marginals are well-specified; a bad marginal confounds copula diagnosis.
- The Gaussian copula underestimates joint extremes; for risk applications use a t-copula or Archimedean copulas (Clayton, Gumbel).
- Empirical copulas are good exploratory tools; fit parametric copulas only after inspecting the pseudo-observation scatter.
- Multivariate copulas beyond dimension 3 are often built via vine copulas (`VineCopula` package); direct multivariate Archimedean copulas scale poorly.
- Copulas capture dependence, not marginal distributions; do not confuse copula correlation with Pearson correlation of the raw variables.

## For Reviewers

**What to look for in a paper using this method.**

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

## See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots